

Suricata Integration with Wazuh

Network Intrusion Detection through SIEM and IDS/IPS Integration

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Domain: SIEM Administration & Network Intrusion Detection

Tools Used: Wazuh SIEM, Suricata IDS/IPS, Ubuntu Endpoint

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1. Introduction

Network-based attacks such as port scans, reconnaissance, and exploitation attempts often leave detectable traces in network traffic before they escalate into a full compromise. This project demonstrates the integration of Suricata, an open-source Intrusion Detection/Prevention System (IDS/IPS) and Network Security Monitoring (NSM) engine, with Wazuh, a SIEM platform used for centralized log collection, correlation, and alerting. By forwarding Suricata's alert log to Wazuh, security events detected at the network level become visible, searchable, and correlated alongside host-based telemetry on a single dashboard.

1.1 Objectives

- Deploy a Wazuh manager/dashboard and register an Ubuntu endpoint as a monitored agent.
- Install and configure Suricata with the Emerging Threats open ruleset on the Ubuntu endpoint.
- Configure the Wazuh agent to ingest Suricata's eve.json alert log.
- Simulate a network reconnaissance attack (Nmap scan) to trigger Suricata detections.
- Visualize and analyze the resulting alerts in the Wazuh Threat Hunting module.

1.2 Scope

The project is implemented in a lab environment consisting of a Wazuh manager/dashboard node and an Ubuntu 20.04 LTS endpoint running both the Wazuh agent and Suricata. Detection is demonstrated using an Nmap service/version scan launched from an external attacker machine (Kali Linux) against the Ubuntu endpoint. The scope covers network intrusion detection and log correlation; it does not include automated blocking or active response.

2. Tools and Environment

Component	Details
SIEM Platform	Wazuh (Manager + Dashboard)
Network IDS/IPS	Suricata 6.0.8 (Emerging Threats open ruleset)
Monitored Endpoint	Ubuntu 20.04.4 LTS (agent: ubuntu_agent)
Attacker Machine	Kali Linux (Nmap scan source)
Log Source	/var/log/suricata/eve.json
Configuration Files	suricata.yaml, ossec.conf (Wazuh Agent)

3. Implementation

3.1 Wazuh Dashboard Overview

Wazuh is deployed as the central manager and dashboard. The overview page summarizes connected agents and alert severity across the environment, providing the baseline view before the Suricata integration is added.

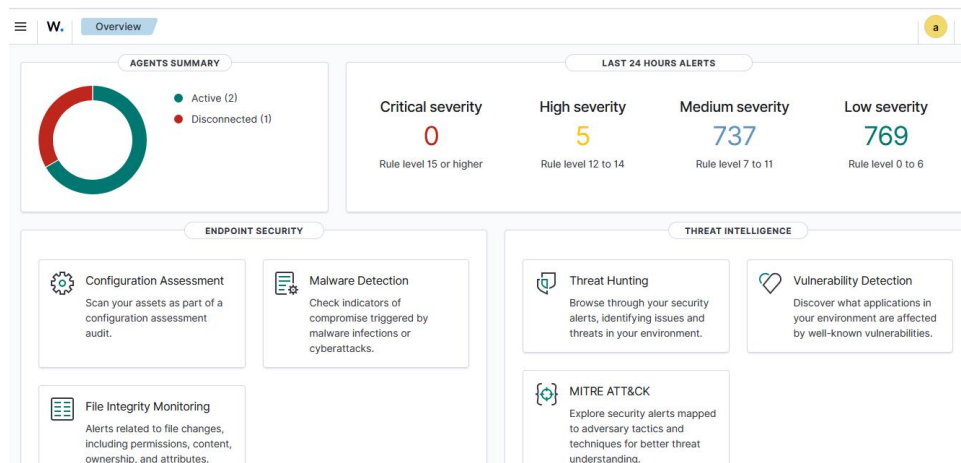


Figure 1: Wazuh dashboard overview showing agent summary and alert severity distribution.

3.2 Installing the Wazuh Agent on Ubuntu

A Wazuh agent is installed and registered on the Ubuntu endpoint. Once active, the endpoint appears in the agents list alongside its IP address, group, and operating system, confirming successful enrollment with the manager.

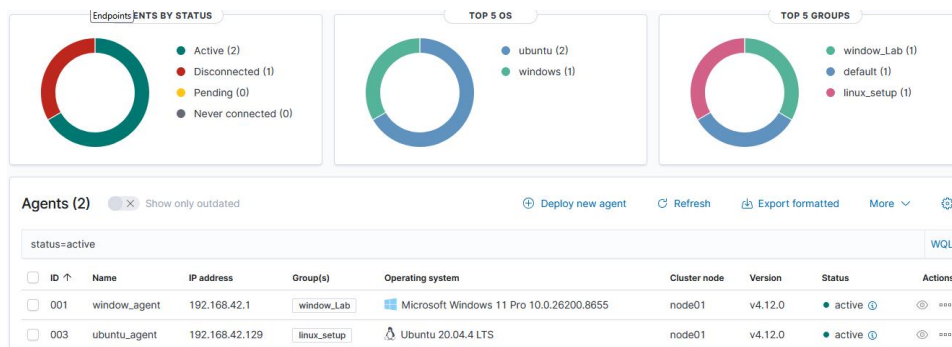


Figure 2: Ubuntu endpoint (ubuntu_agent) registered and shown as active on the Wazuh manager.

```
osboxes@osboxes: ~
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/lib/systemd/system/wazuh-agent.service; enabled; vendor p
   Active: active (running) since Thu 2026-09-17 03:25:02 EDT; 8min ago
   Process: 1100 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (co
   Tasks: 32 (limit: 4539)
   Memory: 737.9M
   CGroup: /system.slice/wazuh-agent.service
           └─1234 /var/ossec/bin/wazuh-execd
             └─1263 /var/ossec/bin/wazuh-agentd
               └─1291 /var/ossec/bin/wazuh-syscheckd
                 └─1304 /var/ossec/bin/wazuh-logcollector
                   └─1311 /var/ossec/bin/wazuh-modulesd

Sep 17 03:24:43 osboxes env[1100]: Deleting PID file '/var/ossec/var/run/wazuh->
Sep 17 03:24:44 osboxes env[1100]: Deleting PID file '/var/ossec/var/run/wazuh->
Sep 17 03:24:44 osboxes env[1100]: Deleting PID file '/var/ossec/var/run/wazuh->
Sep 17 03:24:46 osboxes env[1100]: Started wazuh-execd...
Sep 17 03:24:52 osboxes env[1100]: Started wazuh-agentd...
Sep 17 03:24:54 osboxes env[1100]: Started wazuh-syscheckd...
Sep 17 03:24:57 osboxes env[1100]: Started wazuh-logcollector...
```

Figure 3: wazuh-agent service confirmed running on the Ubuntu endpoint via systemctl.

3.3 Installing Suricata on the Ubuntu Endpoint

Suricata (version 6.0.8) is installed on the Ubuntu endpoint from the official OISF stable PPA:

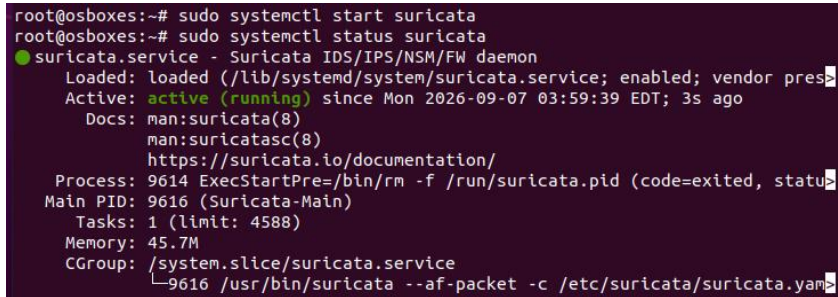
```
sudo add-apt-repository ppa:oisf/suricata-stable
sudo apt-get update
sudo apt-get install suricata -y
```

```
osboxes@osboxes:~$ sudo apt install suricata
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following package was automatically installed and is no longer required:
 gir1.2-goa-1.0
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
 libevent-core-2.1-7 libevent-pthreads-2.1-7 libhiredis0.14 libhyperscan5
 libluajit-5.1-common libnet1 libnetfilter-queue1
The following NEW packages will be installed:
 libevent-core-2.1-7 libevent-pthreads-2.1-7 libhiredis0.14 libhyperscan5
 libluajit-5.1-common libnet1 libnetfilter-queue1 suricata
0 upgraded, 8 newly installed, 0 to remove and 528 not upgraded.
Need to get 6,776 kB of archives.
After this operation, 32.3 MB of additional disk space will be used.
```

Figure 4: Suricata package installation on the Ubuntu endpoint via apt.

After installation, the Suricata service is started and its status is verified to confirm the daemon is active and bound to the monitored network interface:

```
sudo systemctl start suricata
sudo systemctl status suricata
```



```
root@osboxes:~# sudo systemctl start suricata
root@osboxes:~# sudo systemctl status suricata
● suricata.service - Suricata IDS/IPS/NSM/FW daemon
   Loaded: loaded (/lib/systemd/system/suricata.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-09-07 03:59:39 EDT; 3s ago
     Docs: man:suricata(8)
           man:suricata-sc(8)
           https://suricata.io/documentation/
  Process: 9614 ExecStartPre=/bin/rm -f /run/suricata.pid (code=exited, status=0/SUCCESS)
 Main PID: 9616 (Suricata-Main)
    Tasks: 1 (limit: 4588)
   Memory: 45.7M
    CGroup: /system.slice/suricata.service
           └─9616 /usr/bin/suricata --af-packet -c /etc/suricata/suricata.yaml
```

Figure 5: Suricata service running with the af-packet capture interface active.

3.4 Suricata Rules and Configuration

The Emerging Threats open ruleset is downloaded and extracted into Suricata's rules directory so the engine can detect a broad range of known-malicious network activity:

```
cd /tmp/ && curl -LO https://rules.emergingthreats.net/open/suricata-6.0.8/emerging.rules.tar.gz
sudo tar -xvzf emerging.rules.tar.gz && sudo mkdir /etc/suricata/rules
sudo mv rules/*.rules /etc/suricata/rules/
sudo find /etc/suricata/rules -name "*.rules" -exec chmod 777 {} \;
```

Suricata's main configuration file (suricata.yaml) is then updated to define the local network, enable statistics, and bind packet capture to the monitored interface:

```
HOME_NET: "UBUNTU_IP"
EXTERNAL_NET: "any"
default-rule-path: /etc/suricata/rules
rule-files:
  - "*.rules"

# Global stats configuration
stats:
  enabled: yes

# Linux high speed capture support
af-packet:
  - interface: ens33
```

Suricata is restarted to apply the updated configuration: `sudo systemctl restart suricata`

3.5 Wazuh–Suricata Log Integration

To forward Suricata's alert stream to Wazuh, a localfile block is added to the Wazuh agent's `ossec.conf`, pointing to Suricata's JSON event log (`eve.json`):

```
<ossec_config>
  <!--Suricata integration with wazuh-->
  <localfile>
    <log_format>json</log_format>
    <location>/var/log/suricata/eve.json</location>
  </localfile>
</ossec_config>
```

```
<!--Suricata integration with wazuh-->
<localfile>
  <log_format>json</log_format>
  <location>/var/log/suricata/eve.json</location>
</localfile>
/ossec_config>
```

Figure 6: Suricata log-forwarding block added to the Wazuh agent's `ossec.conf`.

The Wazuh agent is restarted to apply the change: `sudo systemctl restart wazuh-agent`

4. Attack Emulation and Detection

With the integration active, Wazuh automatically parses events from `/var/log/suricata/eve.json` and raises corresponding alerts on the dashboard. To validate the pipeline, a service/version detection scan was launched against the Ubuntu endpoint's IP address from an external attacker machine:

```
nmap -sS -sV <ubuntu_ip>
```

```
(kali@kali)-[~]
$ nmap -sS -sV 192.168.42.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-16 07:50 EDT
Nmap scan report for 192.168.42.129
Host is up (0.034s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.41 ((Ubuntu))
MAC Address: 00:0C:29:63:F7:97 (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.92 seconds

(kali@kali)-[~]
$
```

Figure 7: Nmap service/version scan launched against the Ubuntu endpoint from the attacker machine.

4.1 Visualizing the Logs

The resulting alerts can be visualized in the Wazuh dashboard by navigating to the Threat Hunting module and filtering the search bar for events related to the monitored agent.

timestamp	agent.name	rule.description	rule.level	rule.id
Sep 16, 2026 @ 16:50:35.6...	ubuntu_agent	Suricata: Alert - ET SCAN Nmap Scripting Engine User-Agent Detected (Nmap Scripting...	3	86601
Sep 16, 2026 @ 16:50:35.5...	ubuntu_agent	Suricata: Alert - ET SCAN Nmap Scripting Engine User-Agent Detected (Nmap Scripting...	3	86601
Sep 16, 2026 @ 16:50:35.4...	ubuntu_agent	Suricata: Alert - ET SCAN Nmap Scripting Engine User-Agent Detected (Nmap Scripting...	3	86601
Sep 16, 2026 @ 16:50:34.4...	ubuntu_agent	Suricata: Alert - ET SCAN Nmap Scripting Engine User-Agent Detected (Nmap Scripting...	3	86601
Sep 16, 2026 @ 16:50:23.8...	ubuntu_agent	Suricata: Alert - ET SCAN Suspicious inbound to PostgreSQL port 5432	3	86601
Sep 16, 2026 @ 16:50:23.7...	ubuntu_agent	Suricata: Alert - ET SCAN Potential VNC Scan 5800-5820	3	86601
Sep 16, 2026 @ 16:50:23.0...	ubuntu_agent	Suricata: Alert - ET SCAN Suspicious inbound to MSSQL port 1433	3	86601
Sep 16, 2026 @ 16:50:23.0...	ubuntu_agent	Suricata: Alert - ET SCAN Suspicious inbound to Oracle SQL port 1521	3	86601
Sep 16, 2026 @ 16:50:21.6...	ubuntu_agent	Suricata: Alert - ET SCAN Suspicious inbound to MySQL port 3306	3	86601
Sep 16, 2026 @ 16:48:10.3...	ubuntu_agent	SCA summary: CIS Ubuntu Linux 20.04 LTS Benchmark v2.0.0: Score less than 50% (40)	7	19004
Sep 16, 2026 @ 16:48:01.9...	ubuntu_agent	CIS Ubuntu Linux 20.04 LTS Benchmark v2.0.0: Ensure iptables packages are installed: ...	9	19014

Figure 8: Threat Hunting module showing Suricata-generated alerts for the Ubuntu endpoint, including Nmap scan detections and suspicious port probes.

5. Log Analysis

Individual alerts were opened in the Document Details view to inspect the full set of fields Wazuh extracted from the Suricata event, including the rule classification, network 5-tuple, and HTTP metadata associated with the Nmap scripting engine's probe request.

Document Details		View surrounding documents	View single document
Table	JSON		
_index	wazuh-alerts-4.x-2026.09.16		
agent.id	003		
agent.ip	192.168.42.129		
agent.name	ubuntu_agent		
data.alert.action	allowed		
data.alert.category	Web Application Attack		
data.alert.gid	1		
data.alert.metadata.confidence	High		
data.alert.metadata.created_at	2010_07_30		
data.alert.metadata.signature_severity	Informational		
data.alert.metadata.updated_at	2020_04_22		
data.alert.rev	6		

Figure 9: Alert detail showing the Suricata rule classification, category (Web Application Attack), and source agent.

Document Details		View surrounding documents	View single document
data.alert.severity	1		
data.alert.signature	ET SCAN Nmap Scripting Engine User-Agent Detected (Nmap Scripting Engine)		
data.alert.signature_id	2009358		
data.app_proto	http		
data.community_id	1:hMTc15S4/aS9E94DXlnb3RgLjBg=		
data.dest_ip	192.168.42.129		
data.dest_port	80		
data.direction	to_server		
data.event_type	alert		
data.flow.bytes_toclient	728		
data.flow.bytes_toserver	434		
data.flow.dest_ip	192.168.42.129		
data.flow.dest_port	80		
data.flow.pkts_toclient	4		
data.flow.pkts_toserver	4		
data.flow.src_ip	192.168.42.130		

Figure 10: Network flow details including source/destination IPs, ports, byte counts, and the Suricata signature ID for the Nmap scripting engine detection.

Document Details		View surrounding documents	View single document
data.flow.src_ip	192.168.42.130		
data.flow.src_port	60484		
data.flow.start	2026-09-16T07:50:33.902816-0400		
data.flow_id	499868592912764.000000		
data.http.hostname	192.168.42.129		
data.http.http_content_type	text/html		
data.http.http_method	GET		
data.http.http_user_agent	Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)		
data.http.length	276		
data.http.protocol	HTTP/1.1		
data.http.status	404		
data.http.url	/evox/about		
data.in_iface	ens33		
data.ip_v	4		
data.pkt_src	wire/pcap		
data.proto	TCP		

Figure 11: HTTP request metadata showing the Nmap Scripting Engine user-agent string captured during the scan.

data.timestamp	Sep 16, 2026 @ 16:50:33.947
data.ts_progress	request_complete
data.tx_id	0
decoder.name	json
id	1789559435.2515849
input.type	log
location	/var/log/suricata/eve.json
manager.name	wazuh-server
rule.description	Suricata: Alert - ET SCAN Nmap Scripting Engine User-Agent Detected (Nmap Scripting Engine)
rule.firedtimes	8
rule.groups	ids, suricata
rule.id	86601
rule.level	3
rule.mail	false
timestamp	Sep 16, 2026 @ 16:50:35.542

Figure 12: Final rule metadata confirming the alert was decoded from /var/log/suricata/eve.json and classified under the ids/suricata rule group.

5.1 Key Findings

- Suricata successfully detected the Nmap scan using the Emerging Threats signature “ET SCAN Nmap Scripting Engine User-Agent Detected” (signature ID 2009358).
- Wazuh correctly ingested and parsed the Suricata JSON log, decoding it into structured fields such as data.alert.category, data.flow.*, and data.http.*.
- Additional probe attempts were also flagged, including suspicious inbound connections to PostgreSQL, MSSQL, Oracle, MySQL, and VNC ports — indicating the ruleset detects broader reconnaissance behavior, not just the Nmap user-agent signature.
- Alerts were classified at rule level 3 (informational/low severity), which is expected for reconnaissance activity as opposed to a confirmed exploit or compromise.
- Each alert retained full network context (source/destination IP and port, protocol, byte counts), giving analysts enough detail to investigate the source of the scan.

6. Conclusion

The Suricata and Wazuh integration provides an effective way to extend a SIEM's visibility from host-based telemetry into network-level threat detection. Suricata inspects live network traffic against a continuously updated open-source ruleset, while Wazuh centralizes, parses, and correlates the resulting alerts alongside other security data sources. The Nmap scan emulation confirmed that the pipeline detects reconnaissance activity end-to-end, from packet capture on the Ubuntu endpoint to a structured, searchable alert in the Wazuh dashboard.

Overall, this project demonstrates how a network IDS/IPS and a SIEM can be combined to give a security team early warning of reconnaissance and scanning activity often the first observable stage of an attack well before an attacker reaches deeper stages of the kill chain.